MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 8245

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Tetrode, ceramic-metal
See figure 1
Mounting position: Any
Weight: 4 ounces (113.4 grams) nominal

ABSOLUTE RATINGS: F (CW) = 500 MHz, F (pulsed) = 1,500 MHz

Parameter: / Unit:	Ef V ac (Note 1)	eb kv (Note 2)	Eb V dc	ec1 v (Note 2)	Ec1 V dc	ec2 kv (Note 2)	Ec2 V dc	Ib mA dc	ib a (Note 2)	ik a
Maximum:										
Class C Telep:	6.0 ±5%	---	1,500	---	-250	---	300	200	---	---
Class C Teleg:	6.0 ±5%	---	2,000	---	-250	---	300	250	---	---
Class AB:	6.0 ±5%	---	2,000	---	---	---	400	250	---	---
Class C Pulsed:	6.0 ±5%	7.0	---	-500	---	1.5	---	---	6.0	7.0
TEST CONDITIONS:	6.0	---	1,000	---	Adj	---	300	150	---	---

ABSOLUTE RATINGS: F (CW) = 500 MHz, F (pulsed) = 1,500 MHz

Parameter: / Unit:	Pg1 W	Pg2 W	Pp W	Pi W	tp µs (Note 2)	tk s (min)	T(anode core and seal) °C	Cooling --- Note 3
Maximum:								
Class C Telep:	2	12	165	300	---	30	250	---
Class C Teleg:	2	12	250	500	---	30	250	---
Class AB:	2	12	250	500	---	30	250	---
Class C Pulsed:	2	12	250	500	5.0	30	250	---
TEST CONDITIONS:	---	---	---	---	---	120	---	Note 4

GENERAL:

Qualification - Required

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Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 1	5					
1301	Heater current			If	2.30	3.00	A ac
1261	Electrode voltage (grid)			Ec1	-32	-45	V dc
1256	Electrode current (screen)			Ic2	-5.0	+3.0	mA dc
1266	Total grid current	6	Eb = 2,000 V dc; Ec1/Ib = 125 mA dc	Ic1	---	-15	µA dc
1266	Primary grid emission (control)		Ic1 = 70 mA dc; t = 15; anode and g2 floating	Isg1	---	-25	µA dc
1266	Primary grid emission (screen)		Ec1 = 0; t = 15; Ic2 = 100 mA dc; anode floating	Isg2	---	-250	µA dc
2212	Pulse emission (1)	7	Eb = Ec2 = 250 V dc; Ec1 = -100 V dc; prr = 11 ±1; tp = 4,500 µs (min); ec1/ik = 1.5 a; Ef = 5.4 V ac	Δik	---	200	ma
1372	Current division (long pulse, method A)		Eb = Ec2 = 250 V dc; Ec1 = -100 V dc; prr = 11 ±1; tp = 4,500 µs (min); ec1/ib = 1.0 a	egk ic ic2	8.0 --- ---	18.0 200 260	v ma ma
	Quality conformance inspection, part 2						
1031	Low-frequency vibration		No voltages	---	---	---	---
1042	Shock, specified pulse		No voltages; 15 G peak (min); D = 11 ±2 ms; half-sine wave	---	---	---	---
2214	RF useful power output	8	Class C amplifier; F = 470 to 500 MHz; Eb = 2,000 V dc; Ec1 = -90 V dc; Ec2 = 250 to 300 V dc; Ic1 = 25 mA dc (max); Eg1/Ib = 250 mA dc; Ef = 5.50 V ac	Po	225	---	W (useful)
2212	Pulse emission (2)	7	Eb = Ec2 = 250 V dc; Ec1 = -100 V dc; prr = 11 ±1; tp = 4,500 µs (min); ec1/ik = 1.5 a	Δik	---	100	ma
1331	Direct-interelectrode capacitance		Grounded grid	Cpk Cin Cout	--- 14.5 4.2	0.01 19.0 5.2	pF pF pF
1331	Direct-interelectrode capacitance		Grounded cathode	Cgp Cin Cout	--- 25.0 4.2	0.05 29.0 5.2	pF pF pF

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Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3						
---	Life test (1)		Group C; Class C amplifier; F = 470 to 500 MHz; Eb = 2,000 V dc; Ec1 = -90 V dc; Ec2 = 250 to 300 V dc; Ic1 = 25 mA dc (max); Eg1/Ib = 250 mA dc; Ef = 5.50 V ac	t	500	---	hrs
---	Life-test (1) end points:						
2212	Pulse emission (2)	7	Eb = Ec2 = 250 V dc; Ec1 = -100 V dc; prr = 11 ±1; tp = 4,500 µs (min); ec1/ik = 1.5 a	Δik	---	100	ma
1266	Primary grid emission (control)		Ic1 = 70 mA dc; t = 15; anode and g2 floating	Isg1	---	-100	µA dc
1266	Primary grid emission (screen)		Ec1 = 0; t = 15; Ic2 = 100 mA dc; anode floating	Isg2	---	-250	µA dc
---	Life test (2)		Group C; Ec1 = Ec2 = Eb = 0; Ef = 6.6 V ac	t	500	---	hrs
---	Life-test (2) end point:						
1366	Interelement leakage resistance, cold		Rs = 2.5 Meg; supply voltage = E = 100 V dc; g1 negative	Rg1k	10	---	Meg
			E = 500 V dc; g2 positive	Rg2K	10	---	Meg
			E = 500 V dc; g1 negative	Rg1g2	10	---	Meg
---	Pulse power output	9	eb = 7.0 kv; ec2 = 1.2 kv; ib = 6.0 a; tp = 5 µs; prr = 1,000; F = 1,200 ±50 MHz	po	17	---	kw

NOTES:

1. At frequencies above approximately 300 MHz it may be necessary to reduce heater voltage to compensate for rf transit-time heating of the cathode after dynamic operation of the tube has started. This back heating is a function of frequency, grid current, grid bias, anode current, duty cycle, and circuit design and adjustment. There is an optimum heater voltage which will maintain the cathode at the correct operating temperature for any particular set of operating conditions. A maximum variation of ±5 percent from optimum shall be maintained. For straight-through, class C CW amplifier operation, the following heater operation voltages are indicated:

Frequency (MHz)	Ef (V ac)
301 to 400	5.75
401 to 500	5.50

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2. With a pulse repetition rate (prr) of 1,000 or less and a pulse duration (tp) of 5 µs or less, peak anode current (ib) shall be limited to 6.0 amperes. For pulses of longer duration or higher repetition rates, peak anode current shall be reduced in accordance wih the data shown on figure 2, and pulse anode voltage, screen voltage, and grid voltage shall be reduced in accordance with the data shown on figure 3.

3. Forced-air cooling shall be provided for the base and anode. The maximum seal and anode core temperature rating shall not be exceeded. At an anode dissipation of 250 watts and with an incoming air temperature of 25°C maximum at sea level, a minimum airflow of 3.8 cfm shall pass through the anode cooler. With no voltages applied to the tube, the static pressure drop across the anode cooler with an airflow of 3.8 cfm (at 25°C at sea level) is approximately 0.15 inch of water. The pressure drop varies with the amount of escaping air and with the shape and construction of the air director. This airflow requirement applies at bias voltages less than -100 volts and at frequencies less than 500 MHz (CW) and 1,500 MHz (pulsed). Air cooling on the tube shall be increased with increased negative grid bias, increased incoming air temperature, or increased frequency of operation. The stem shall be cooled either by a portion of the anode air or by a separate air supply. Airflow shall be applied before or simultaneously with the electrode voltages and may be removed simultaneously with them. In cases where long life and reliable operation are factors, operation with cooling in excess of minimum requirements is recommended.

4. In all electrical tests involving application of heater voltage, forced-air cooling is permitted at the rate of 4.0 cfm maximum for the base and anode. A separate source may be used for the base and anode but neither source shall exceed 4.0 cfm. The cooling air shall not have a temperature less than 20°C nor an absolute pressure greater than 32 in. Hg.

5. The AQL for each test in quality conformance inspection, part 1, shall be 0.65, inspection level II.

6. This test shall be the first test performed at the conclusion of the holding period.

7. The input wave shape shall have a tr and a tf of 25 µs maximum each, and the slope of the top of the pulse may be not greater than 0.5 percent with a ripple not to exceed 0.1 percent.

8. Use cavity and circuit as shown on Drawing 224-JAN.

9. This test shall be performed during the initial production and once each succeeding 12-calendar month period in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes having an acceptance number of zero, and a second sample of three tubes having a combined acceptance number of one. In the event of failure, the test shall be made as a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

10. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
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Review activities:
Navy - MC
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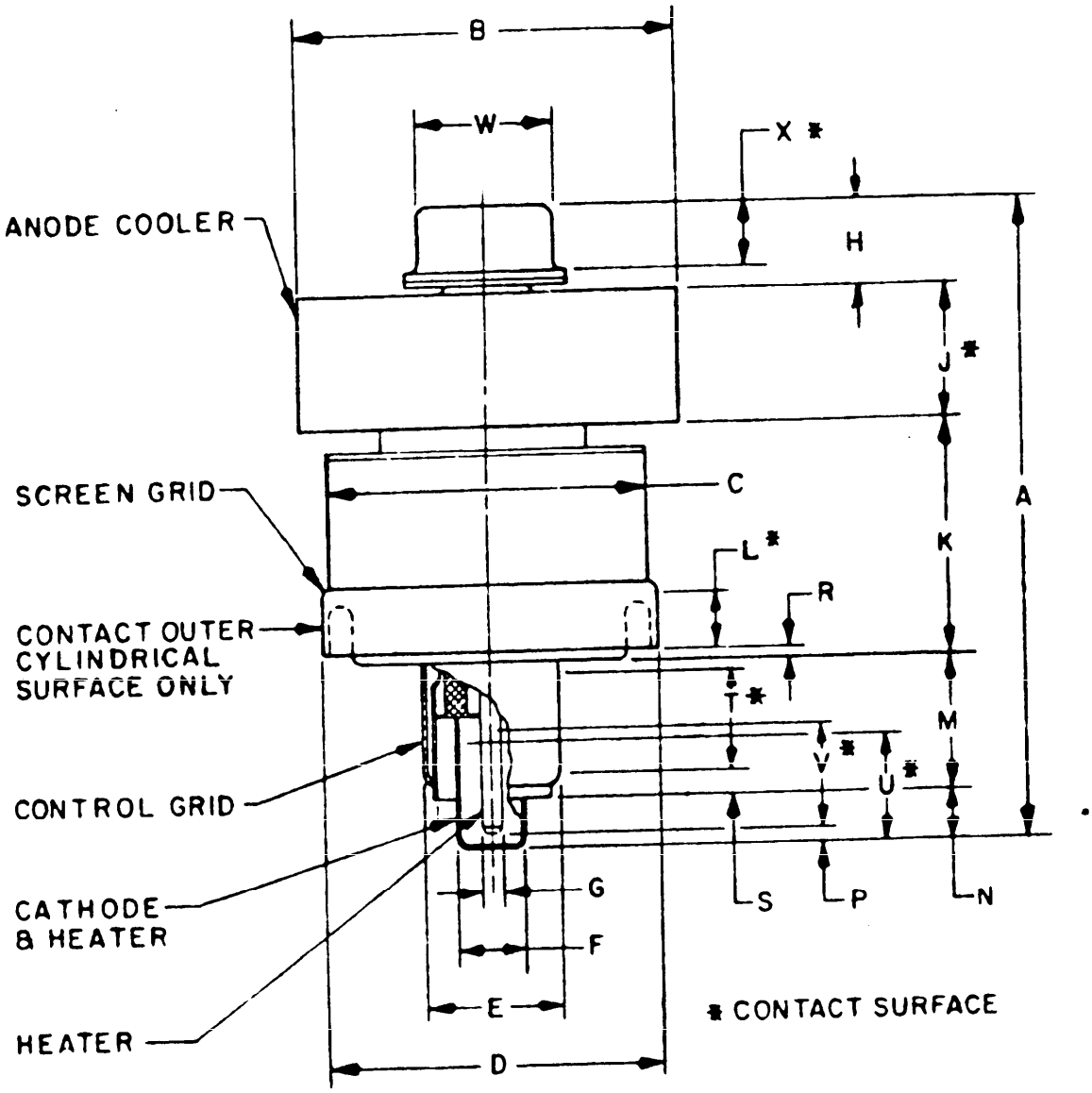


FIGURE 1. Outline drawing of electron tube type 8245.

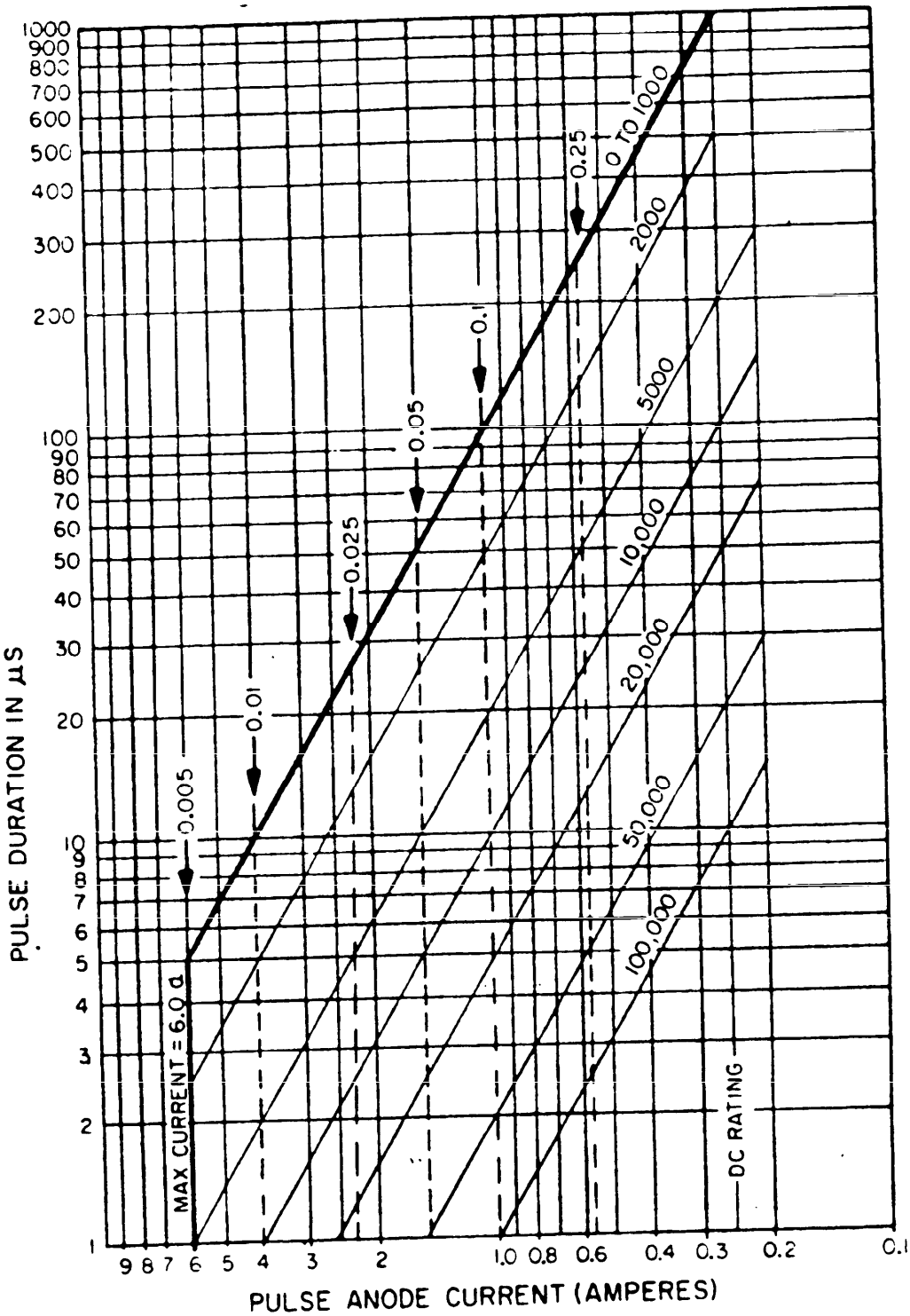
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A		2.813 (71.45)
B	1.610 (40.89)	1.640 (41.66)
D	1.410 (35.81)	1.440 (36.58)
E	.587 (14.91)	.597 (15.16)
F	.317 (8.05)	.327 (8.31)
G	.088 (2.24)	.098 (2.49)
Quality conformance inspection, part 3 (see note)		
C		1.406 (35.71)
H		.358 (9.09)
J	.710 (18.03)	.790 (20.07)
K	.740 (18.80)	.820 (20.83)
L	.187 (4.75)	
M	.500 (12.70)	.580 (14.73)
N	.235 (5.97)	.265 (6.73)
P	.032 (.81)	.062 (1.57)
T	.344 (8.74)	
U	.406 (10.31)	
V	.468 (11.89)	
W	.559 (14.20)	.573 (14.55)
X	.240 (6.10)	
Reference dimensions		
R	.020 (.51)	
S	.125 (3.18)	

NOTE: Dimensions shall be checked yearly. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, AQL of 6.5, inspection level S3. The regular yearly double sampling plan may be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 8245 - Continued.

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NOTES:

a. Solid lines represent constant repetition rates; dashed lines represent constant duty factors.

b. DO NOT EXTRAPOLATE ABOVE OR TO THE LEFT OF BOLD LINES.

FIGURE 2. Anode current versus pulse duration.

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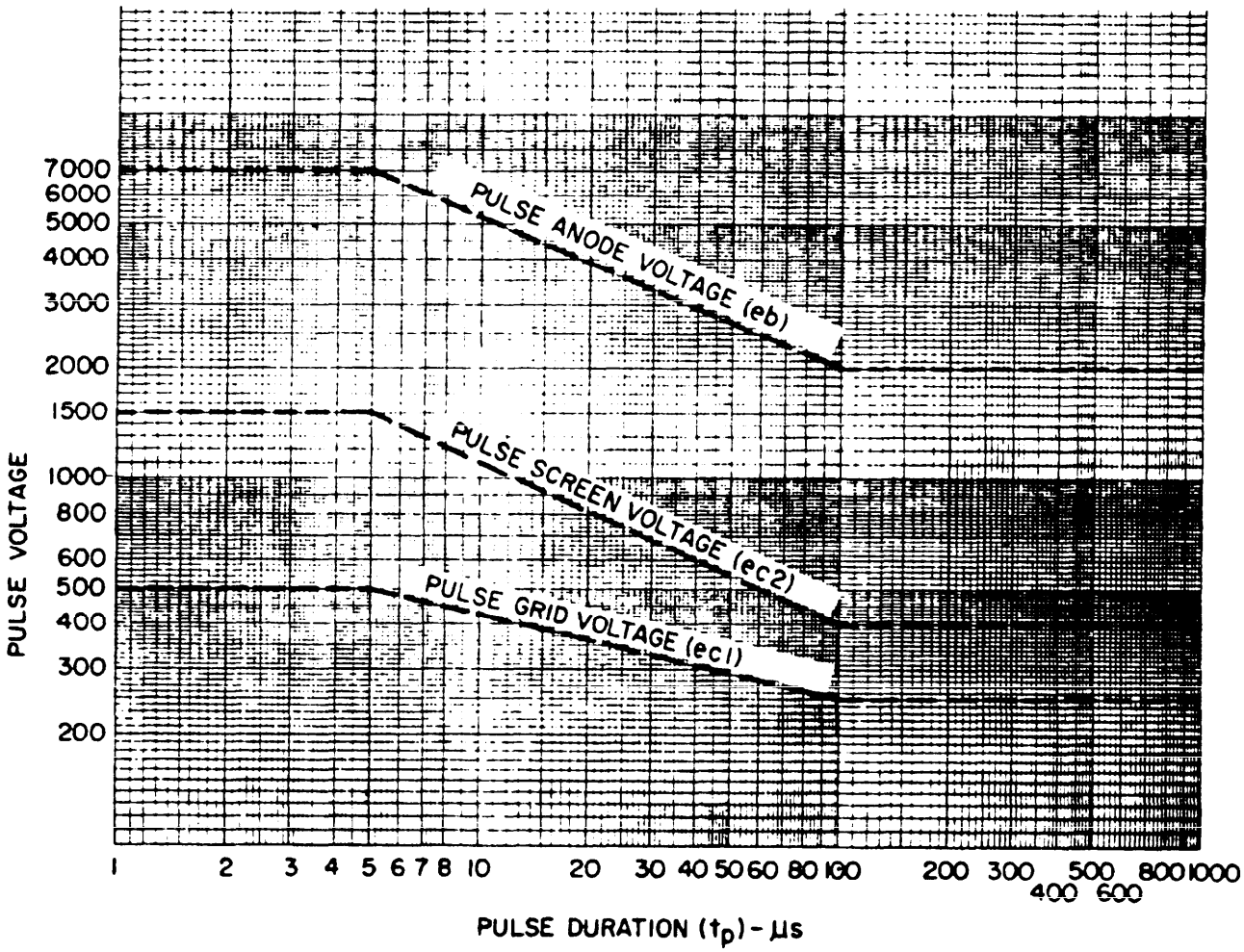


FIGURE 3. Derating chart: pulse anode voltage, screen voltage and grid voltage vs. pulse duration (tp).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-1/1506C	Electron Tube, Power, Type 8245

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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